An Investigation of Differential Siblicide strategies in Masked and Blue Footed Booby birds

Introduction

Sibling rivalry is a common ecological phenomenon among avians, in which individuals of the same brood will compete for resources and parental care (Krebs and Davies, 1993). In its most extreme cases, this escalates to siblicide, where one sibling causes the other to perish– often by direct confrontation or eviction of the opposing chick from the nest (Krebs and Davies, 1993, Anderson, 1989). Siblicide itself also comes in multiple forms, as seen in Blue Footed Booby (BFB– *Sula nebouxii*) and Masked Booby (MB– *Sula dactylatra*) seabirds: both may lay 1-3 eggs, but exhibit differential frequencies of parent-permitted siblicide in which only one chick survives to fledge (Anderson, 1989).

BFBs and MBs both occupy parts of the tropical eastern Pacific Ocean, with overlapping ranges from the Gulf of California to Peru (Harris, 2001, Mansoor, 2004). Both species hunt fish to feed their young, and exhibit similar diet and lifestyles (Harris, 2001, Mansoor, 2004). Although closely related, these birds differ significantly in brood-rearing (Harris, 2001, Mansoor, 2004). While BFBs are facultatively siblicidal, parents only tolerating it during periods of food shortage, MBs exhibit obligate siblicide where only one chick of each MB brood typically survives (Anderson 1989, Harris, 2001, Mansoor, 2004).

Although seemingly counterintuitive under Hamilton's famed kin selection theory, siblicide in MBs and BFBs can be favored under natural selection if it means obtaining more resources and increasing one's own chances of survival and reproduction (Eberhard, 1975). Still, despite the advantage conferred to the dominant/'winning' sibling, there exists a divide in the research concerning the reason why parents permit siblicide. The early death of any progeny would appear to present a huge indirect loss of fitness for the parent (Anderson, 1989). How does siblicide arise in these species, and what factors contribute to this divergence in parental permissivity of siblicide?

The dominant hypothesis explaining this phenomenon is that it is most resource-efficient to only rear one/few offspring (Lougheed and Anderson, 1999). Production of excess offspring/eggs serve as "insurance", increasing the chances that at least one egg hatches and reaches reproductive age– this may be especially necessary in boobies, as it is not uncommon for booby eggs to fail to hatch (Anderson, 1989). This hypothesis is supported by the phenomenon

of asynchronous brooding, in which eggs laid together hatch at different times (Anderson, 1989). In MBs, the first-hatched "A" individual eliminates their "B" sibling shortly after their late emergence. Hatching asynchrony both increases survivors per nest and reduces the food a parent must procure for their brood per surviving chick per day, conferring a direct parental benefit and indicating that brood size may be limited by parental ability to provide for all offspring (Lougheed and Anderson, 1999). This model suggests that the siblicidal dichotomy between MBs and BFBs is caused by differential permissivity driven by resource availability, and predicts that an inherent restriction of MB ability to efficiently procure resources for multiple young may drive their obligately siblicidal tendencies. In contrast, BFBs may be able to more effectively provide for multiple young, meaning that the production of excess offspring benefits BFBs in volatile environments by allowing them to rear more offspring in favorable resource conditions (Drummond and Rodriguez, 2024, Barraza et. al., 2024).

Still, other evidence contends that resource limitation cannot explain the dichotomy of siblicide between MB and BFBs (Anderson and Ricklefs, 1992). One study documented that preventing MB siblicide and forcing double brood conditions caused more offspring per brood to survive to fledging– indicating that MBs are obligately siblicidal despite the parent's optimal brood size exceeding one (Anderson, 1989). Further investigation concluded that both species have the ability to successfully provide sufficient food to support double broods (Anderson and Ricklefs, 1992) . This challenges the notion that differential siblicide strategies in MBs and BFBs are mediated by resource procurement limitations, giving rise to a parent-offspring conflict model in which siblicide is driven by offspring behavior outside of parental behavioral control and interests. The model predicts that unlike MBs, facultatively siblicidal BFB offspring must have less incentive to engage in siblicide, allowing for agreement between parental and sibling ideal brood sizes (Anderson and Ricklefs, 1992).

A 2008 study further complicated this debate by showing that Brown Boobies, which have similar siblicidal patterns to MBs, exhibit aggression-aggression relationships between nestlings as well as equal allocation of androgenic hormones and yolk to all eggs in a clutch (Drummond et. al., 2008). In contrast, BFBs exhibited low-aggression dominance-submission relationships between offspring and declining yolk masses with hatch order, suggesting that B offspring were predisposed towards a subordinate role from the womb (Drummond et. al., 2008). These findings suggest that an interplay between parental influence and sibling competition

mediates the development of a sibling hierarchy or siblicide (Drummond and Canales, 1997). Cross-fostering studies also indicate that both parental and offspring species contribute to the siblicidal outcomes of a brood, further supporting the idea of a collaborative relationship between offspring predisposition to aggression and parental permissivity in determining siblicide strategy in these birds (Lougheed and Anderson, 1999).

Although much research has been conducted in an attempt to understand the factors that mediate the differences between BFB and MB siblicidal tendencies, the current literature has fatal gaps that fail to explain why BFBs are facultatively siblicidal in response to resource availability while MBs exhibit obligate siblicide even when it appears to be favorable for parents to rear multiple offspring. The central goal of this project is to investigate the selective pressures and physiological/genetic mechanisms that have led to the different siblicide strategies in MBs and BFBs. Furthermore, I will seek to determine the extent to which parental behaviors and nestling strategies contribute to this observed variation.

My main hypothesis is that differential offspring conflict strategy drives the siblicide dichotomy between MBs and BFBs. My rationale for this hypothesis is that brood siblicide outcomes do not appear to be controlled by resource restrictions, and obligate MB siblicide appears to clash with the ability of MB parents to provide for more than one chick. Therefore, obligate siblicide may be a nestling strategy maintained in MBs through the evolutionary advantage that single-chick status confers for MB offspring. Conversely, in BFBs, a familial hierarchy system replaces direct sibling competition, allowing for the presence of double broods in resource-permitting conditions. My secondary hypothesis is that the differential competitive strategies of MB and BFB offspring facilitate the reproductive interests of their respective parents over the lifetime, rather than representing a parent-offspring-conflict relationship. My reasoning for this is that brood siblicide is determined both by parent and offspring species, with MB parents increasing the rates of BFB chick siblicide and BFB parents decreasing the rates of MB chick siblicide. This indicates that the status of obligate or facultative siblicide in booby species is enforced by parents and offspring, and therefore that this siblicidal arrangement presents evolutionary benefits for both parties.

Research Aims

Aim 1: Determine the role of differential strategies of sibling competition in the siblicidal trends of Masked and Blue Footed Boobies.

Hypothesis 1: Double brooded MB nestlings will engage in comparatively more physical confrontation over food in comparison with double brooded BFBs, which will adopt a consistent and non-aggressive pattern of hierarchy-driven unequal food allocation between chicks.

Rationale: Previous research has demonstrated that obligately siblicidal species engage in relentless aggression while BFBs are predisposed to sibling hierarchies which discourage offspring aggression. I expect that my findings pertaining to food and physical conflict, a major determinant of nestling fitness and survival, will align with prior research concerning these general trends.

Methods: This study will take place from January to May at the mixed-breeding MB and BFB colony in Punta Cevallos, on the Isla Espaniola in the Galapagos, which is a part of the natural range of both MB and BFBs. 15 MB nests will be experimentally doubled via the addition of an unrelated but similarly sized and aged first-hatching chick from a separate brood to the next within 7 days of the original chick's hatching. 15 double BFB nests will be selected from the population. Using inconspicuous video surveillance, nests will be monitored for chick eviction. Evicted chicks will be placed back into nests until chicks reach 10 days of age, at which point they typically will be able to fend for themselves. The same covert video surveillance will allow for the observation of nestling aggression and confrontation over food. Daily food intake will be measured by weighing chicks at 4 hour intervals on a 24 hour cycle, using the benchmark of an 5% increase in body mass as an indicator of feeding in the previous 4 hours. The sum of body mass increases will provide an estimate of the total food intake over each 24-hour cycle. Measurements will continue until birds reach 50 days of age or fledge. Other than necessary interference, there will be as little alteration of the habitat as possible.

Expected Results: I expect that MBs in the forced double brood condition will engage in more food-related physical confrontations that mediate nestling feeding frequency/food intake, while BFBs will engage in fewer food-related physical confrontations and instead follow a pattern of food intake following a consistent sibling hierarchy. Significant results would suggest the existence of differential nestling strategies in MBs and BFBs in double brood situations, necessitating the existence of an adaptive reason for the presence of these divergent strategies.

Hypothesis 2: Double-brooded MBs will exhibit significantly lower reproductive success than their single-brood counterparts, while double versus single-brood status of dominant BFBs will not be a significant predictor of reproductive success.

Rationale: Although previous research has demonstrated that MBs have the ability to support double broods to fledging, there is little exploration of the effects of brooding size on lifetime offspring fitness. If the optimal strategy of the MB is to eliminate a sibling, then single-chick status will confer an observable fitness/reproductive advantage over the lifetime. Similarly, if the optimal strategy of the dominant BFB does not involve the elimination of a sibling, then single-chick status will not confer the same advantage as seen in MBs.

Methods: This experiment will take place on the same reserve as described under hypothesis 1, and development of double brooded MB and BFB nests will mirror the procedure described above as well. For single-chick broods, 15 nests of each species will be identified, and after the hatching of the first chick, any other eggs present will be removed from the nest. First-born/dominant birds will be covertly tagged for identification and tracked throughout the study. Each year, birds will be monitored during rearing season (typically January-May) for survival and number of offspring successfully fledged. Measurements will continue until the end of each bird's life span. Life span and amount of offspring successfully fledged will be compared between single and double brooded populations of each species.

Expected Results: I anticipate that while brood status will be a predictor of lifetime fitness and reproductive success in MBs, it will not bear significance on BFB fitness or success. Such results would indicate that preventing MBs from engaging in their typical obligate siblicide strategy

reduces MB fitness, while BFBs who are not obstructed from engaging in their typical sibling hierarchy strategy by being in a double brood will not have reduced fitness. This would demonstrate that BFB and MB sibling competition strategies do bear a significant effect on lifetime fitness, therefore providing an evolutionary reason for the nestling-driven differential development of obligate and facultative siblicide in MBs and BFBs.

Aim 2: Explore the evolutionary dynamics and biological underpinnings of differential siblicide permissivity in Blue Footed Boobies and Masked Boobies.

Hypothesis 3: MBs who are forced to rear double broods will experience diminished future fitness/reproductive success when compared to single-brood controls, while double brooding BFBs will not.

Rationale: Despite the ability of both MBs and BFBs to rear double broods to fledging, MB parenthood increases brood siblicide, while BFB parenthood does not. Therefore, I expect that rather than a direct benefit to the current brood, the obligate siblicide strategy of MBs may confer fitness advantages for MBs in the future or when raising subsequent broods.

Method: Location, single and double brood development, and observation will mirror the procedures described above. Both female and male parental birds will be tagged and tracked throughout the study. Each year, birds will be monitored during rearing season for survival and the number of offspring successfully fledged. Measurements will continue until the end of each bird's life span. Life span and amount of offspring successfully fledged will be compared between single and double brooded populations of each species.

Expected Results: I expect that double-brooding MBs will have shorter life spans and/or rear fewer future offspring when compared to siblicide-allowing controls, while this difference will not be seen in BFBs. These results would support the notion that obligate siblicide in MBs presents an evolutionary benefit that may outweigh the cost of losing one nestling to siblicide, while BFBs experience no such evolutionary benefit– therefore suggesting that offspring siblicidal strategy in these birds is complementary to parental evolutionary interests.

Hypothesis 4: MBs and BFBs will exhibit differential progesterone receptor (PR) expression in the brain, specifically in regions pertinent to parental behavior like the hypothalamus, hippocampus, and amygdala.

Rationale: Progesterone is a hormone implicated in parenting, and PR KO results in altered parental behavior (Schneider et. al., 2003). Therefore, if evolutionary differences in parental siblicide permissivity and rearing strategy are genetically encoded, they may manifest in differences in PR expression and localization between MB and BFB brains.

Methods: The brains of three males and three females of each species (12 individuals total) from the Punta Cevallos reserve population will be used for the production of audiograms depicting PR expression. The method for PR staining will follow the one used in Insel et. al. (1992) for OTR staining. Brains will remain frozen on dry ice at -70°C until they are made into cryostat-cut 20 μm slide-mounted sections, which will be used for the binding assay. 125I radiolabeled progesterone will be used along with a high-affinity synthetic antiprogestin, mifepristone, for the definition of nonspecific binding. When dry, slides will be apposed to Hyperfilm along with 125I plastic standards for 2-3 days. Using the Macintosh IMAGE program, autoradiograms will be analyzed by using a third-order polynomial regression. Nonspecific bindings will be subtracted from total binding to reveal specific binding values. Anatomic localization will be defined by counterstaining postfixed sections with thionin and overlaying computer images of stained sections and autoradiograms.

Expected results: I anticipate that MBs and BFBs will exhibit differential PR expression and localization in brain regions pertinent to parental behavior. Significant results would point to the existence of neurobiologically-encoded differences in siblicide permissivity, supporting the hypothesis that parental behavior and biology enforce the siblicidal dichotomy between MBs and BFBs.

Conclusion

MBs and BFBs are birds that, despite close relatedness and similarities in behavior, habitat, and diet, exhibit two distinct phenotypes of sibling competition: obligate and facultative siblicide. Although this phenomenon is well-documented across a number of environmental conditions, the current literature fails to come to a unified understanding of the origins and mechanism of this dichotomy, representing a deeper lapse in our understanding of siblicide and sibling competition itself. Thus, there is great value in research elucidating the evolutionary and biological underpinnings of this behavior.

The proposed project seeks to develop a unifying theory of MB and BFB siblicide that accounts for evolutionary pressures and behavioral strategies from both parents and offspring as well as illuminates possible proximate and biological mechanisms behind siblicide in these birds. By taking into account long-term fitness considerations of both parents and offspring, and comparing the strategies of BFBs and MBs, this project expands upon the classical approach towards the study of siblicide in booby populations while integrating multiple theories of evolutionary siblicide strategy. Additionally, mapping PR expression in the brains of these birds will provide a novel perspective on the biological basis of this behavior and lay the foundation for further research on the mechanisms through which siblicide is exerted and permitted in boobies.

Furthermore, the ability of BFBs to use facultative siblicide as a tool for maximum exploitation of environmental resources in volatile conditions is both impressive and significant for conservation efforts of tropical birds and other species amidst the human-driven rapid change in global climate (Bizberg-Barraza et. al., 2024). This project will reveal a deeper understanding of the factors that drive this plasticity in BFBs in comparison to MBs, possibly holding greater significance in understanding the vulnerability and resistance of at-risk populations to changes in environment and resources the world around.

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